



important information

- Diuretics should not be given in Acute Congestive Heart Failure
- ↓ tachycardia (compensatory mechanism) → Worsens CHF

ON EXAMINATION

FRAMINGHAM CRITERIA

LEFT VENTRICULAR FAILURE	RIGHT VENTRICULAR FAILURE
• Tachycardia (early sign) ◦ Maximum HR = $220 - \text{age}$ • SBP ↓, Oliguria • S₃ present - Ventricular gallop rhythm • B/L crackles/crepitations ◦ Cause- ↑ parenchymal blood vessel (of lungs) hydrostatic pressure → Fluid leaks into alveoli → Pulmonary edema  • B/L ↓ air entry ◦ Cause- ↑ pleural blood vessel hydrostatic pressure → Pleural effusion • ↓ vital capacity	• M/c cause of right ventricular failure → Left ventricle failure • Tachycardia (early sign) ◦ Maximum HR = $220 - \text{age}$ • JVP ↑ ◦ Congested neck veins ◦ Kussmaul's sign-Paradoxical rise of JVP on inspiration ◦ Abdomino- jugular reflex positive • Tender hepatomegaly • Ascites • B/L Ankle pitting edema } Longer duration of development

CRITERIA

MINOR CRITERIA - (Mnemonic - PET-HAND)

- **P**leural effusion (B/L, transudative)
- **T**achycardia (HR > 120/ min)
- **H**epatomegaly, **A**nkle edema
- **N**octurnal cough
- **D**yspnea on (ordinary) exertion

MAJOR CRITERIA

- **E**verything except minor criteria like B/L crackles, ↑ JVP

- For diagnosis - 2 major criteria / 1 major + 2 minor criteria



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	No structural damage, risk factors present
STAGE B	Structural damage, no signs/ symptoms of CHF
STAGE C	Structural damage, signs/ symptoms of CHF
STAGE D	Refractory CHF → Cardiac transplantation (from brain-dead donor)

KILLIP CLASSIFICATION

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I	No features of CHF
II	Mild CHF findings - Rales, S3, JVP ↑
III	Acute pulmonary edema
IV	Cardiogenic shock (SBP ↓, Urine output ↓, Pulmonary edema)

WORK-UP

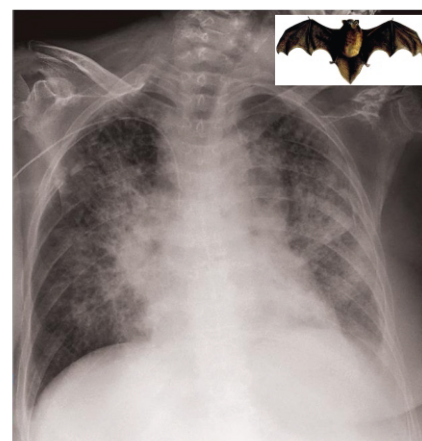
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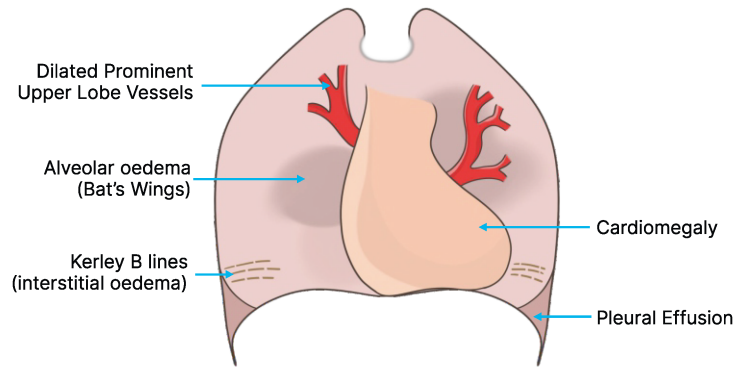
1. N-terminal Pro BNP (B type natriuretic peptide)

- Biomarker for early diagnosis
- Released from ventricles of heart
 - In response to volume loading
- Causes natriuresis
- Differentiates cardiogenic pulmonary edema (values elevated) from non-cardiogenic pulmonary edema (eg: ARDS (values normal))

2. Chest X ray/ Radiological finding

- **Earliest** radiological finding - Prominent upper lobe veins/Antler sign /Reverse mustache sign
- Bat wing pulmonary edema
 - Congestion in cardiac blood vessels → ↑ hydrostatic pressure (in blood vessels) → leakage of fluid out of blood vessels → development of whitish lung fields
- ↑CT ratio (> 0.5) - Cardiomegaly
- Blunting of costophrenic angle
 - Cause - B/L pleural effusion
- Kerley B lines - Perpendicular to pleural surface
 - Congested distal lung blood vessels
 - Also seen in lymphangitis carcinomatosa





3. Echocardiography

- For accurate assessment of ventricular size, function, valve morphology
 - Informs about ejection fraction

- Ejection fraction = $\frac{\text{Stroke volume}}{\text{End diastolic volume}} = 65\%$

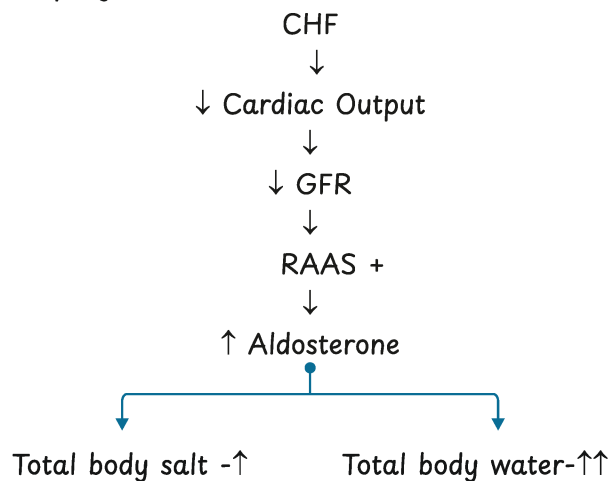
- EF < 40% - Heart failure with reduced ejection fraction (HFrEF)
- EF > 50% - Heart failure with preserved ejection fraction (HFpEF)

4. Cardiac MRI

- **Imaging of choice** (in Heart Failure)
- Assessment of -
 - Ventricular size assessment
 - Etiology, Progression of HF

5. Serum electrolytes

- Na⁺ concentration ↓ - Negative prognostic indicator (at time of admission in HF patient)



- K concentration ↓ - Due to diuretic usage

6. New biomarkers for CHF

- Galectin-3



ACUTE CONGESTIVE HEART FAILURE

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CONDITION	ACUTE PULMONARY EDEMA	CARDIOGENIC SHOCK
MANIFESTATION	- Breathing difficulty - Crackles - Pink frothy sputum	- BP↓ - Urine output ↓ (Renal failure) - Pulmonary congestion
T/t	Lasix - Furosemide, Torsemide Morphine-↓ pulmonary edema (by shunting of blood from pulmonary to systemic circulation) Nitrates (NTG drip) - or Nesiritide (natriuresis) - or Nitroprusside-Acute onset HTN (cause LVF) O2 (via Non-Invasive Ventilation) - Indications - - Acute pulmonary edema - Acute exacerbation of chronic bronchitis Positioning - Head higher than feet ↓ venous return to heart - New drugs - - Serelaxin, Ularitide	A. Inotropes - - Dobutamine - β 1 agonist - Levosimendan - - Calcium sensitizer → ↑ responsiveness of troponin (to Ca²⁺ levels) - Affects potassium channels of blood vessels → vasodilation - Milrinone (Ionodilator) B. Mechanical circulatory support - - IABP-Intra-aortic balloon/ counter pulsation pump ↑ coronary blood flow - Inflates during diastole

CHRONIC CONGESTIVE HEART FAILURE

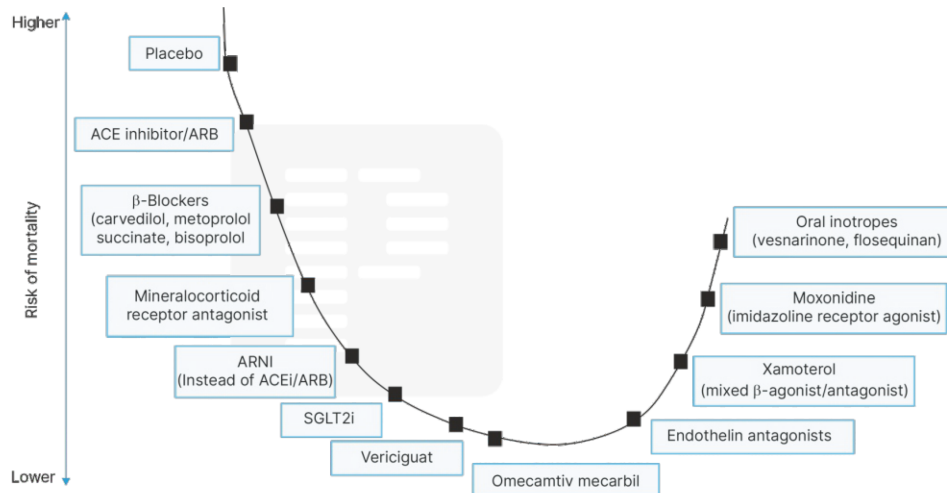
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CONDITION	HFrEF (Heart Failure with reduced Ejection Fraction)	HFpEF (Heart Failure with preserved Ejection Fraction (> 50%))
CAUSES	- Coronary artery disease - Acute HTN - Valve lesions - Cardiomyopathy	- HTN - Radiation - Ageing - Restrictive Cardiomyopathy - Endocardial fibroelastosis
T/t	- ACE inhibitors/ ARB - Maximum reduction in mortality - β blockers (Carvedilol, Metoprolol, Bisoprolol) - Spironolactone - Angiotensin Receptor - Neprilysin Inhibitor (ARNI) (If ACE inhibitors can't be given)	- ARNI (may be beneficial) - Control of HTN - Spironolactone (may be beneficial)- Control of HTN - SGLT2 inhibitors (under investigation) - Control of HTN - Renin-angiotensin-aldosterone-directed therapy



- Valsartan (angiotensin receptor blocker) - Blocks angiotensin II → prevents vasoconstriction
 - +
 - Sacubitril (neprilysin inhibitor) -inhibits breakdown of BNP → ↑ BNP action → Natriuresis
 - 5. SGLT2 (Sodium Glucose Linked Transporter 2) inhibitor - Empagliflozin
 - Causes urinary sugar loss
 - Causes salt loss → ↓ mortality
 - 6. Vericiguat- ↑ cGMP
 - 7. Omecamtiv
 - Cardiac myosin activator
- ACE inhibitors & ARB - ineffective
 - Aldosterone antagonists (maybe beneficial)
 - 5. Digoxin
 - 6. Ineffective β blockers & calcium channel blockers
 - 7. Phosphodiesterase 3 inhibitors
 - Sildenafil → ineffective

MORTALITY BENEFIT (HFrEF)



MEDICALLY REFRACTORY CONGESTIVE HEART FAILURE

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- **T/t -**
- 1. LVAD (left ventricular assist device)
- 2. CRT (Cardiac Resynchronization Therapy)
 - Done in Dilated Cardiomyopathy
- 3. Cardiac transplantation

Important Information

- **Important Information**
- Metoprolol (β blocker) C/I in -
 - Acute Heart Failure / Acute Pulmonary Edema
 - Symptomatic Bradycardia (HR < 60/min) (causing hypotension)
 - Asthma / COPD
- Grading of Dyspnea -



HA Class		Level of Clinical Impairment
I		No Limitation of physical activity. Ordinary physical activity does not cause undue breathlessness, fatigue, or palpitations.
II		Slight Limitation of physical activity. Comfortable at rest, but ordinary physical activity results in undue breathlessness, fatigue, or palpitations.
III		Marked Limitation of physical activity. Comfortable at rest, but less than ordinary physical activity results in breathlessness, fatigue, or palpitations.
IV		Unable to carry on any physical activity without discomfort. Symptoms at rest can be present. If any physical activity is undertaken, discomfort is increased.



Q. Which of the following drugs will reduce the progression of chronic congestive heart failure in your patient?

- a. ACEI
- b. Beta blocker
- c. CCB
- d. Diuretics

Ans. (a)

Q. Which of the following will not be seen in case of acute heart failure?

- a. Pulsus alternans
- b. Reverse bat wing appearance
- c. Distended neck veins
- d. Nocturia

Ans. (b)

Q. A 50-year-old lady with HTN is brought with shortness of breath and dyspnea on minimal exertion for the past 2 days. Physical examination reveals the presence of elevated JVP and crepitations in both the lung fields along with palpable liver 3 cm below the costal margin. Which of the following is not correct about this case?

- a. Increased PCWP
- b. Narrow split S2
- c. Start on beta blockers
- d. Start intravenous nitroglycerin

Ans. (c)



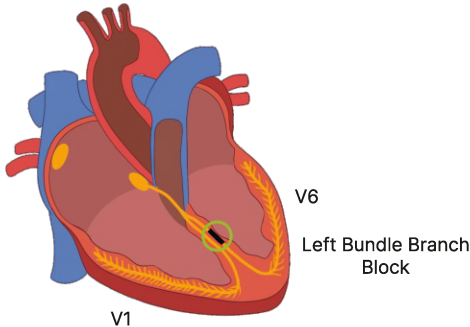
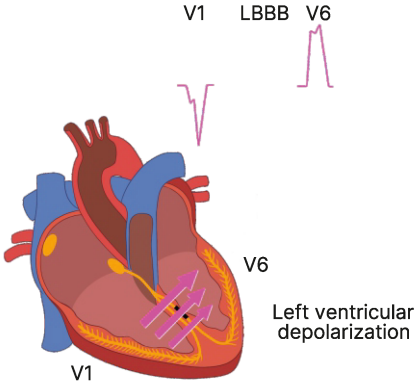
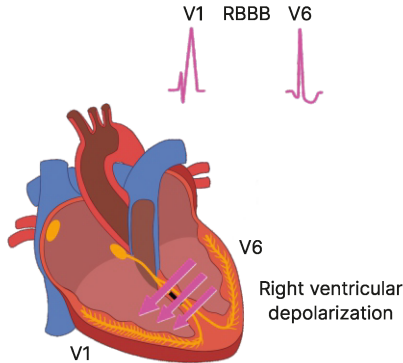
BUNDLE BRANCH BLOCK

BUNDLE BRANCH BLOCK

PATHOLOGY

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- In normal individual - Simultaneous contraction of ventricles
- In diseased individual - Sequential contraction of ventricles

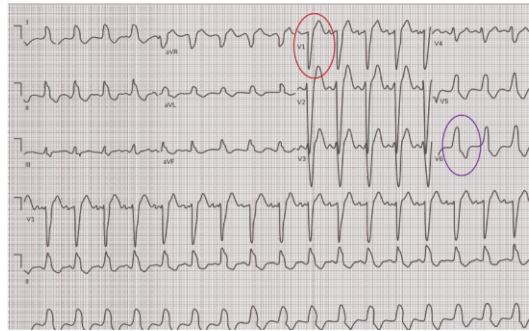
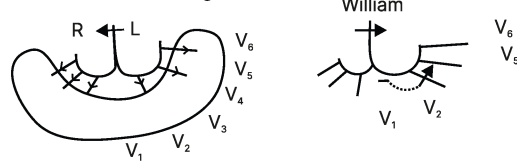
	LEFT BUNDLE BRANCH BLOCK (LBBB)	RIGHT BUNDLE BRANCH BLOCK (RBBB)
PATHOPHYSIOLOGY	• Right ventricle contracts and sends blood to lungs • Left ventricle - Delayed contraction • Pooling of blood in lungs → Pulmonary edema  V6 Left Bundle Branch Block  V1 LBBB V6 Left ventricular depolarization	• Left ventricle - activated normally • Right ventricle activation delayed • Tricuspid and pulmonary valve contracts late  V1 RBBB V6 Right ventricular depolarization
CAUSES	Mnemonic- Kim Has my Heart blocked • Hyperkalemia • Ischemic Heart Disease • Myocardial Infarction • Hypertension • Aortic stenosis	Mnemonic - COR2E • Cor pulmonale • Right ventricular hypertrophy • Pulmonary embolism • Physiological

- S1 - Wide split
- S2 - Wide split

Mnemonic - WiLLiaM

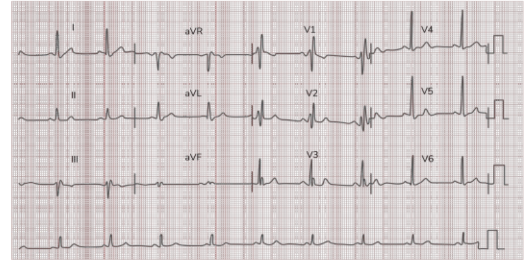
- QRS > 120 ms/Broad QRS
- V1 & V2 - Dominant S wave
 - Rabbit ear appearance (W) - Seen in terminal part of S-wave
- V5 & V6 - Broad R wave
 - Rabbit ear appearance (M) - Seen in terminal part of R-wave
- Absent Q wave
- ST segment changes

ECG findings in LBBB



Mnemonic - MarroW

- V1 & V2 - rSR' pattern (M pattern)
 - r - Left ventricular activation
 - R' - Right ventricular activation
- V5 & V6 -
 - Broad/Dominant S wave
 - Terminal part - W pattern



Important Information

- Bundle branch blocks → Current does not pass through conduction systems
 - Passes via muscles → **Intermyocyte Conduction**



Q. Which of the following causes left bundle branch block?

- a. Hyperkalemia
- b. Hypokalemia
- c. Hyponatremia
- d. Hypernatremia

Ans: (a)

Q. Identify the false statement about the ECG findings in left bundle branch block.

- a. QRS > 120 msec
- b. Absent q wave in lead V5 and V6
- c. ST-segment elevation
- d. rSR in lead V1

Ans: (d)



13. RHEUMATIC HEART DISEASE

SECONDARY PROPHYLAXIS IN RF

00:00:40

- Rheumatic Fever (RF) : caused by Group A β -hemolytic Streptococcus (GABHS)
- All diagnosed RF patients are started on **benzathine penicillin IM every 3-4 weeks** for secondary prophylaxis
- M/c valvular lesions:
 - Children: **Mitral regurgitation**
 - Adults (>18 yrs): **Mitral stenosis**

AHA RECOMMENDATION FOR SECONDARY PROPHYLAXIS

CONDITION	DURATION OF ANTIBIOTICS
RF WITHOUT CARDITIS	• Up to 21 years of age or 5 years after last attack (whichever is longer)
RF WITH CARDITIS, NO VALVULAR DISEASE	• Up to 21 years of age or 10 years after last attack (whichever is longer)
RF WITH PERSISTENT VALVULAR DISEASE evident clinically / by echocardiography	• Up to 40 years of age or 10 years after last attack (whichever is longer)

- Penicillin injections should be given after Test dose d/t risk of anaphylaxis
- In **penicillin allergy**: Macrolides (Azithromycin daily) for the same duration as penicillin

ANAPHYLAXIS RISK WITH PENICILLIN

00:09:48

- Cause of death: **Laryngeal edema**
- Cells involved: **Basophil-mediated histamine release**
- Cytokine involved : **IL-4**
- Intervention:
 - Adrenaline 0.5 mg IM, 1:1000 (undiluted)
 - Repeat after 5 minutes if needed + 1-2 L of saline

PATHOGENESIS OF RHEUMATIC FEVER

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- Mechanism: Molecular mimicry
- Causative agent: **Group A β -hemolytic Streptococcus** (GABHS)
- Antibodies formed against **M protein** cross-react with host connective tissue components
- Cross-reactivity targets: **N-acetylglucosamine** in valvular endothelium and joint synovium, Basal ganglia
- Immune response type: **Type II hypersensitivity reaction**
- Pathognomonic feature: **Aschoff nodules**

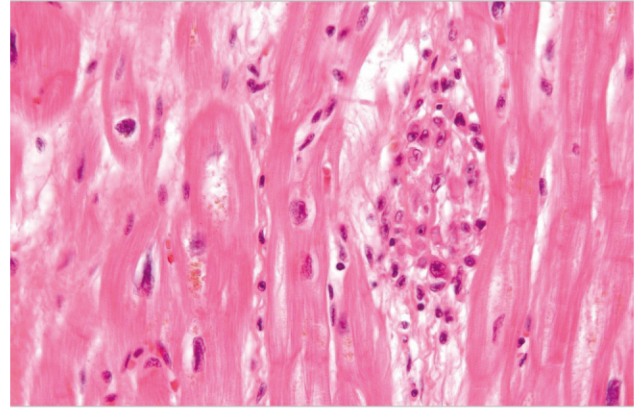


Antisicken cells (caterpillar cells)

• Presents in Aschoff nodules

• They are histiocytes

- Age group: children between 5-14 yrs
- **M/c** cause of acquired heart disease in developing countries



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COURSE OF DISEASE

- Untreated ARF: Disease progression over 12 weeks
- Treated ARF: course modified to 1-2 weeks
- Recurrent sore throats lead to arthritis after **3 weeks**
- Early feature: Arthritis, carditis (**earliest**)
- Late features: Chorea, **indolent carditis**

ACUTE RHEUMATIC FEVER - MAJOR CRITERIA

- Age Group: 5-14 years

Mnemonic: C.A.S.E.S

- **C** - Carditis (60-75% of cases)
- **A** - Arthritis (50-75% of cases)
- **S** - Sydenham's chorea
- **E** - Erythema marginatum
- **S** - Subcutaneous nodules

CARDITIS

- Pancarditis: Involvement of all three layers of the heart (endocardium, myocardium, and pericardium)
- The layers may also be affected individually or sequentially

1. RHEUMATIC PERICARDITIS	• Chest pain at rest • Auscultation: Pericardial Friction Rub • ECG: ST Elevation (concave upwards)
2. RHEUMATIC MYOCARDITIS	• Patient can go into Acute Heart Failure • Management: Mnemonic- LMNOP ○ Lasix ○ Morphine ○ Nitrates ○ Oxygen ○ Positioning
3. ENDOCARDITIS	• Characteristic manifestation



VALVE INVOLVEMENT

- M/c: **Mitral > Aortic**
- Rare: **Pulmonic Valve**
 - RHD commonly leads to Mitral stenosis
 - Long standing Mitral stenosis cause PAH
 - PAH may result in mild Pulmonic Regurgitation d/t elevated pulmonary artery pressure

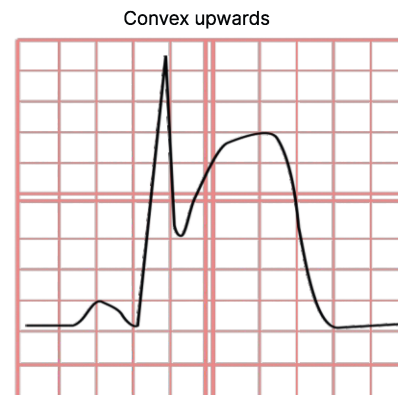
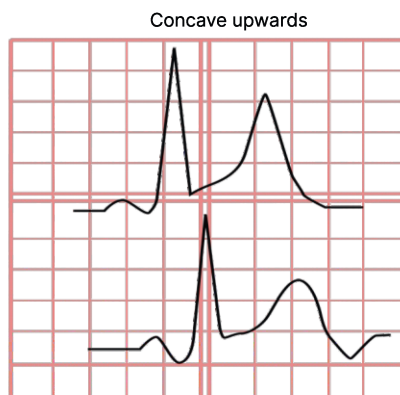
LESION

- In Child/Active carditis: **Mitral Regurgitation**
- In Adult/Carditis: **Mitral Stenosis**

SURFACE INVOLVEMENT

- Commissural end valvulitis
- Valvulitis → Inflammation of cusp of the valve → Leakage of blood → **Carey Coombs murmur**
- Major diagnostic criteria for RF: **Echocardiographic evidence of valvulitis**

ECG



- ST elevation Concave upwards: **Acute pericarditis**
- ST elevation Convex upwards: **MI**

ARTHRITIS (RHEUMATIC ARTHRITIS)

00:37:23

- Affects Large Joints: Ankle, Knee, Elbow joint
- Low-risk population (e.g., Americans)
 - < 2/100,000 school-age children
 - < 1/1000 population
 - Classical presentation: **Migratory Polyarthritis**
- Moderate - High-risk population (e.g., India):
 - Monoarthritis, Polyarthralgia

SYDENHAM CHOREA

00:41:05

- Basal ganglia involved: **Caudate nucleus**
- Fast distal involuntary movement, semi-purposive
- Incidence: **Females > males**
- Chorea Gravidarum: Recurrence of chorea at the time of pregnancy



- Prolonged latent period after sore throat (late feature)
- ASO titer: **Normal**
- Arthritis (50%)

CLINICAL FEATURES

- Darting tongue / Harlequin tongue
- Pronator sign
- Hung up reflexes
- Emotional lability

Important Information

- Hung up ankle jerk → associated with ↓ T4 (hypothyroidism)

Important Information

ATHETOSIS VS CHOREA

- Athetosis: Slow, distal involuntary movement, often seen in cerebral palsy
- Chorea: Fast, distal involuntary movement



SUBCUTANEOUS NODULES

- Typically found over the
 - Extensor aspect of fingers
 - Olecranon process
 - Occipital region
 - Tip of spinous process of vertebra



Subcutaneous nodules

ERYTHEMA MARGINATUM

- Rare in Indian population
- Characterized by **Serpiginous margins** with **central clearing**

Erythema marginatum

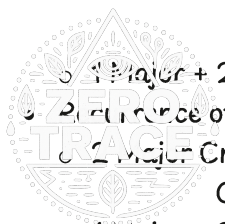


MODIFIED JONES CRITERIA 2015 UPDATE

ALGORITHM

- Initial Acute Rheumatic Fever
 - 2 Major Criteria
- OR

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- 1 Major + 2 Minor Criteria
- Absence of Acute Rheumatic Fever:
 - 2 Major Criteria
 - OR
 - 1 Major + 2 Minor Criteria
 - OR
 - 3 Minor Criteria

MAJOR CRITERIA

LOW-RISK POPULATION	MODERATE - HIGH-RISK POPULATION
Carditis: Clinical, Echo	Same
Migratory Polyarthritits	Monoarthritis, Polyarthralgia
Chorea	Same
Subcutaneous Nodules	Same
Erythema Marginatum Rash (EM rash)	Same

MINOR CRITERIA

LOW-RISK POPULATION	MODERATE - HIGH-RISK POPULATION
Fever > 38.5°C	Fever > 38°C
ESR > 60 mm in 1st hour	ESR > 30 mm in 1st hour
CRP > 3 mg/dL	CRP > 3 mg/dL
PR Interval Prolonged / 1st degree Heart Block	PR Interval Prolonged / 1st degree Heart Block
Polyarthralgia	Monoarthralgia

BLOOD WORK / LABS

00:56:21

- Always Requested Investigations for Diagnosis of Rheumatic Fever:
 - ECG
 - ECHO
 - CBC
 - Streptococcal serology :ASO Titer ↑, Anti-DNase B ↑
- Tests to Exclude Alternate Diagnoses:
 - Autoimmune panel
 - Viral serology
 - Urine test for Chlamydia

MANAGEMENT OF THE DISEASE

00:58:34

- Primary prevention with antibiotics
 - Oral Amoxicillin if onset of sore throat is < 9 days
- Aspirin
 - Used for arthritis
 - Watch for:
 - Salicylate toxicity
 - Reye syndrome
- Steroids
 - Used for pericarditis
 - Controversial use
- Acute Heart Failure (HF): **LMNOP**
 - **L**asix
 - **M**orphine
 - **N**itrates
 - **O**xygen
 - **P**ositioning

CHOREA MANAGEMENT

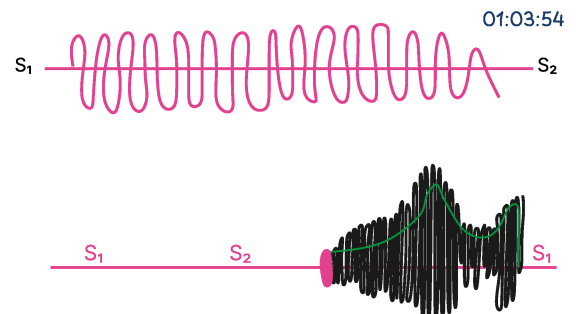
- Generally self-limiting
- Mild cases: Calm environment is sufficient
- Moderate to Severe cases:
 - Medications:
 - Carbamazepine
 - Valproate
 - Haloperidol: S/e of drug induced parkinsonism
- Severe cases: **Steroids**
- Medically refractory chorea: **IVIG** (Intravenous Immunoglobulin)

Important Information

- IVIG also used in management of:
 - Kawasaki disease
 - Guillain-Barré Syndrome (GBS)

ECHO EVIDENCE OF RHEUMATIC CARDITIS

- Pathological MR: Blood leaking from LV→LA
 - Jet length >2 cm
 - Peak velocity >3 m/sec
 - Chronic MR → **Pansystolic murmur**
- Pathological AR:
 - Jet length >1 cm
 - Peak velocity >3 m/sec
 - **Mid diastolic murmur with a presystolic accentuation**
- MS (Mitral Stenosis): LA minus LV gradient >4 mm Hg





- Q. 18-year-old female is having pain in left wrist joint. On examination a murmur is heard. Work up shows increased ESR and CRP. Diagnosis is?
- a. Rheumatoid arthritis
 - b. Rheumatic fever
 - c. Ankylosing spondylitis
 - d. Juvenile idiopathic arthritis

Ans. (b)

- Q. An 11-year-old child with a history of recurrent streptococcal pharyngitis presents to you with recurrence of fever and polyarthralgia. There is no features of carditis or valvular disease. How often is benzathine penicillin i.m recommended in secondary prevention of this case?
- a. Single dose once
 - b. Thrice weekly lifelong basis
 - c. Once every 4 weeks till the age of 21 years
 - d. Once every 4 weeks till age of 40 year

Ans. (c)



INFECTIVE ENDOCARDITIS

CAUSATIVE ORGANISM

00:02:24

NATIVE VALVE ENDOCARDITIS	- Community acquired: Streptococci - Hospital acquired: Staphylococcus aureus	
PROSTHETIC VALVE ENDOCARDITIS (PVE)	Indication for prosthetic valve deployment: Severe calcified Mitral stenosis	
	DURATION AFTER SURGERY	CAUSATIVE ORGANISM
	<2 months	Coagulase-negative Staphylococci (C.O.N.S)
	2-12 months	C.O.N.S
	>12 months	Streptococcus viridans
TAVR-PVE	Organisms: Enterococci > Staphylococcus aureus	
CIED-IE	- Cardiac Implantable Electronic Device – Infective Endocarditis - Devices: ICD, CRT - Pacemaker shows less incidence of IE as compared to ICD, CRT - Organism: Staphylococcus aureus	
PWID / IV DRUG USERS	- Right-sided Endocarditis - Commonly involves Tricuspid Valve - Causative organism: MRSA - Left-sided Endocarditis - Congenital lesion responsible: Patent Fossa Ovalis (Right to Left shunt) with Tricuspid stenosis - Causative organism: MRSA > Enterococcus, Pseudomonas, Candida	

INCIDENCE OF INFECTIVE ENDOCARDITIS

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- Incidence of IE
 - Incidence of IE is directly proportional to the pressure gradient across heart valves
 - Since the left side of the heart has higher pressure gradients, left-sided valves have a higher risk of developing IE
- Valvular lesion with highest chance of IE: **MR (Mitral Regurgitation)**



MR.	• Systolic event (LV to LA) • Operates at the highest-pressure gradient ◦ LV: 120 mmHg ◦ LA: 4-5 mmHg ◦ Pressure difference is the highest
MS	• Occurs in Diastole → low pressure situation
AR	• Occurs in Diastole → low pressure situation
AS	• Systolic event (LV - Aorta) ◦ LV: 130-140 mmHg d/t Aortic stenosis ◦ Aorta: 120 mmHg ◦ Pressure difference is less

- Least incidence of IE: Atrial Septal Defect
 - LA → RA, Pressure difference of only 4 mmHg

PROGRESSION OF THE DISEASE

- Acute IE:
 - Duration: Days to weeks
 - Presentation: Hectic febrile illness, death
- Subacute IE:
 - Indolent course
 - Worsening of valvular lesion
 - New onset murmur (d/t perforation in valve leaflet)
 - Embolic events

INFECTIVE ENDARTERITIS

- Analogous process where infection develops in:
 1. A-V fistula
 2. AA shunt- PDA (Patent Ductus Arteriosus)
 - Ductus arteriosus connects Aorta to Pulmonary artery

PROSTHETIC VALVE ENDOCARDITIS (PVE)

- More common in <1 year of surgery
- Incidence of IE in **Bioprosthetic valve > Metallic valve**
 - Bioprosthetic: Bovine, Porcine
 - Metallic: Requires life-long anticoagulation



Metallic valve

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CULTURE-NEGATIVE ENDOCARDITIS

CAUSES:

- Prior antibiotic exposure
- Fastidious organisms:
 - Granulicatella



- Abiotrophia
- HACEK group
 - Haemophilus saprophilus
 - Aggregatibacter species
 - Cardiobacterium hominis
 - Eikenella
 - Kingella
- Bartonella
- Tropheryma whipplei
- Coxiella burnetii

DIAGNOSIS TOOLS FOR FASTIDIOUS ORGANISMS

- Shot Metagenomics: Next generation sequencing of DNA of pathogens
- PCR / Amplicon

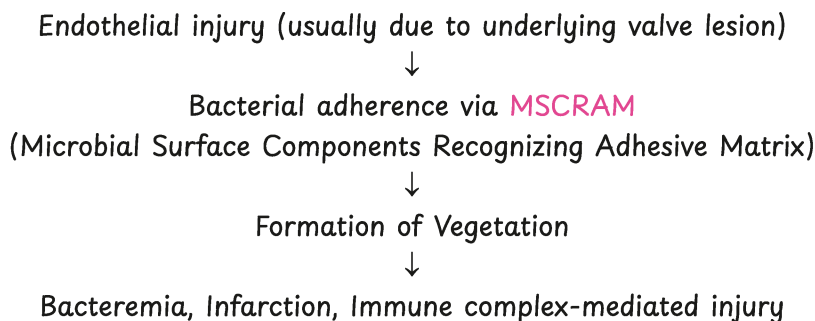
PORTALS OF ENTRY

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- Oral cavity, skin, URT → Streptococcus, Staphylococcus aureus
- GIT (polyps, tumor) → Streptococcus gallolyticus
- Genitourinary → Enterococci

PATHOGENESIS

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EXTRA MILE

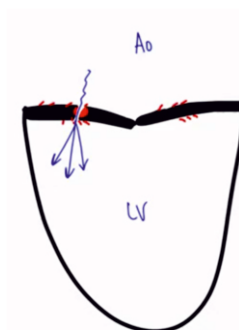
NBTE (Non-Bacterial Thrombotic Endocarditis) / **sterile vegetations**

- Occurs in hypercoagulable states
 - Acute promyelocytic leukemia: **M3 AML**
 - Antiphospholipid antibody syndrome

MANIFESTATIONS

CARDIAC MANIFESTATIONS:

- Perforation: Contributing to **new onset murmur**
- Paravalvular abscess
- Ring abscess
- Heart block
- CHF (Congestive Heart Failure)
- Intracardiac fistula
- Coronary embolization of vegetations



00:35:30